

## Worksheet No. - 03

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**Subject Name:** Advanced Internet Programming Lab **Subject Code:** 25CAP-656

**Aim/Overview of the practical:**(a): use JNDI naming for Java database connectivity.

**Objective:** To understand how to use JNDI naming to configure and access a database connection in a Java application, enabling centralized management of database resources and decoupling the application code from database connection details.

### Procedure/Algorithm/Code

#### NewClass.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
@WebServlet("/students")
```

```
public class NewClass extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {
```

```
        response.setContentType("text/html");
```

```
        PrintWriter out = response.getWriter();
```

```
        out.println("<html>");
```

```
        out.println("<head><title>Employee List</title></head>");
```

```
        out.println("<body>");
```

```
        out.println("<h2>Employee Details</h2>");
```

```
        try {
```

```
            // Load MySQL Driver
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

```
                "jdbc:mysql://localhost:3306/DSS",
```

```
                "root",
```

```
                "abc@123"
```

```
            );
```

```
            Statement stmt = con.createStatement();
```

```
ResultSet rs = stmt.executeQuery("SELECT * FROM EMP;");
```

```
out.println("<table border='1'>");
```

```
out.println("<tr>");
```

```
out.println("<th>No</th>");
```

```
out.println("<th>Name</th>");
```

```
out.println("<th>Age</th>");
```

```
out.println("<th>Title</th>");
```

```
out.println("</tr>");
```

```
while (rs.next()) {
```

```
    out.println("<tr>");
```

```
    out.println("<td>" + rs.getInt("ENo") + "</td>");
```

```
    out.println("<td>" + rs.getString("EName") + "</td>");
```

```
    out.println("<td>" + rs.getInt("EAge") + "</td>");
```

```
    out.println("<td>" + rs.getString("Title") + "</td>");
```

```
    out.println("</tr>");
```

```
}
```

```
out.println("</table>");
```

```
rs.close();
```

```
stmt.close();
```

```
con.close();
```

```
} catch (Exception e) {
```

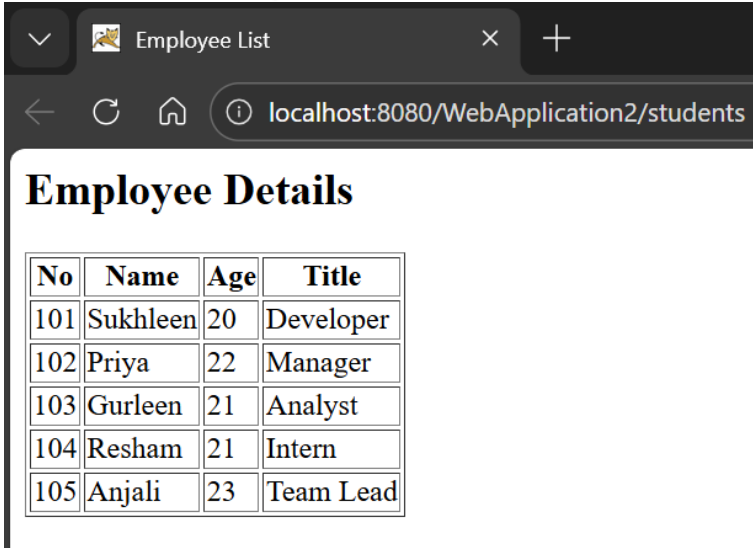
```
        out.println("<p style='color:red;'>Error: " + e.getMessage() + "</p>");  
    }  
  
    out.println("</body>");  
  
    out.println("</html>");  
  
    }  
  
}
```

### SQL Query

```
CREATE DATABASE DSS;  
  
USE DSS;  
  
DROP Table EMP;  
  
CREATE TABLE EMP (  
    ENo VARCHAR(10),  
    EName VARCHAR(50),  
    EAge INT,  
    Title VARCHAR(50)  
);  
  
INSERT INTO EMP VALUES ('101', 'Sukhleen', 20, 'Developer');  
INSERT INTO EMP VALUES ('102', 'Priya', 22, 'Manager');  
INSERT INTO EMP VALUES ('103', 'Gurleen', 21, 'Analyst');  
INSERT INTO EMP VALUES ('104', 'Resham', 21, 'Intern');  
INSERT INTO EMP VALUES ('105', 'Anjali', 23, 'Team Lead');  
  
SELECT *FROM EMP;
```

### Output

(a)



| No  | Name     | Age | Title     |
|-----|----------|-----|-----------|
| 101 | Sukhleen | 20  | Developer |
| 102 | Priya    | 22  | Manager   |
| 103 | Gurleen  | 21  | Analyst   |
| 104 | Resham   | 21  | Intern    |
| 105 | Anjali   | 23  | Team Lead |

**Learning outcomes (What I have learnt):**

1. Configure a JNDI resource for a database in a Java EE environment (e.g., Tomcat).
2. Use JNDI lookup to obtain a database connection in a servlet or Java application.
3. Execute SQL queries using the JNDI-obtained DataSource connection.